# Kafka Data Collector

## Cloud Insights

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August 24, 2020

This PDF was generated from https://docs.netapp.com/us-en/cloudinsights/task_config_telegraf_kafka.html on July 21, 2021. Always check docs.netapp.com for the latest.

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# Kafka Data Collector

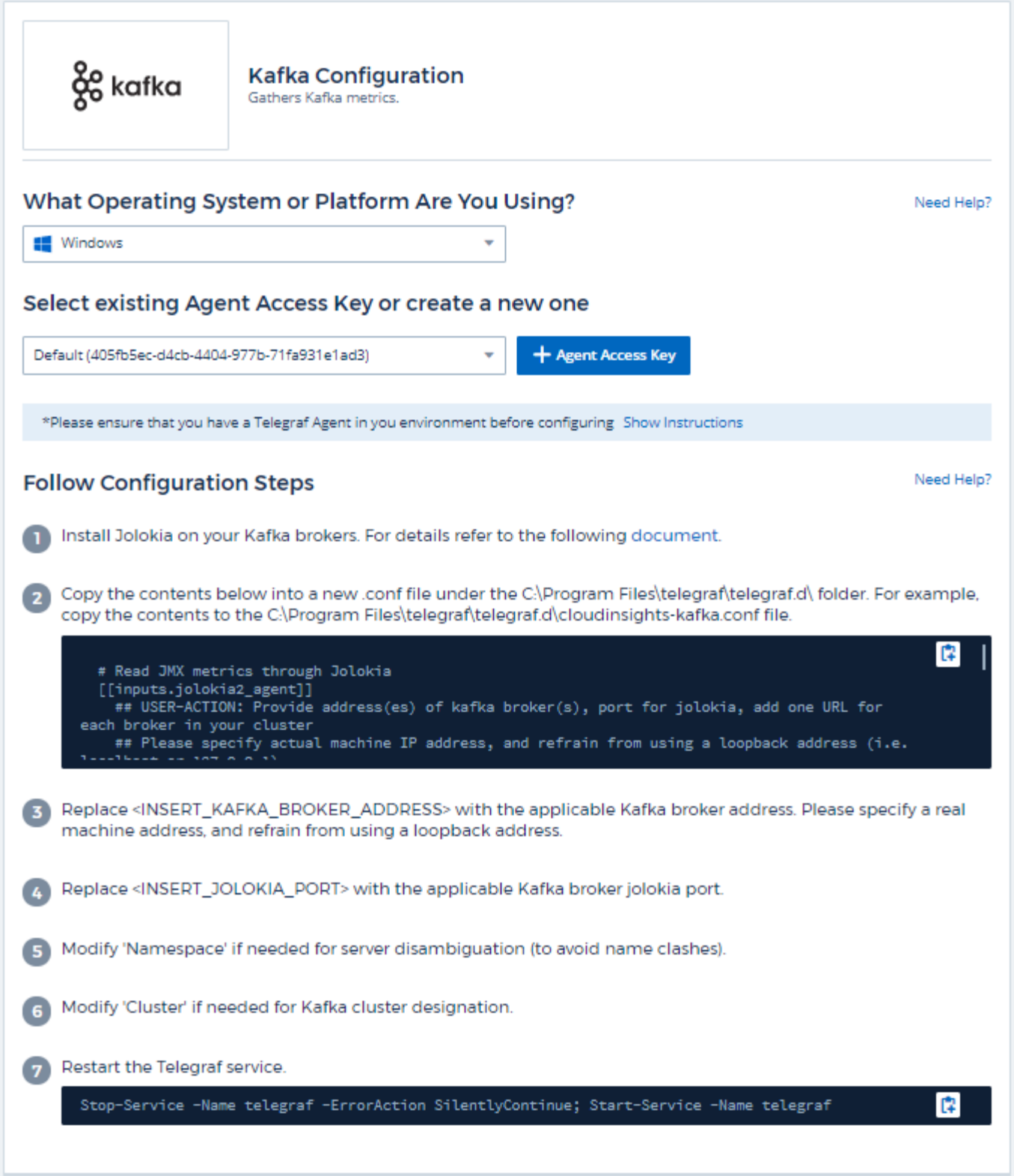
Cloud Insights uses this data collector to gather metrics from Kafka.

## Installation

1. From **Admin > Data Collectors**, click **+Data Collector**. Under **Services**, choose Kafka.

   Select the Operating System or Platform on which the Telegraf agent is installed.

2. If you haven't already installed an Agent for collection, or you wish to install an Agent for a different Operating System or Platform, click *Show Instructions* to expand the Agent installation instructions.
3. Select the Agent Access Key for use with this data collector. You can add a new Agent Access Key by clicking the **+ Agent Access Key** button. Best practice: Use a different Agent Access Key only when you want to group data collectors, for example, by OS/Platform.
4. Follow the configuration steps to configure the data collector. The instructions vary depending on the type of Operating System or Platform you are using to collect data.




# Setup

The Kafka plugin is based on the telegraf's Jolokia plugin. As such as a requirement to gather info from all Kafka brokers, JMX needs to be configured and exposed via Jolokia on all components.

### Compatibility

Configuration was developed against Kafka version 0.11.0.2.

### Setting up

All the instructions below assume your install location for kafka is '/opt/kafka'. You can adapt instructions below to reflect your install location.

### Jolokia Agent Jar

A version the Jolokia agent jar file must be downloaded. The version tested against was Jolokia agent 1.6.0.

Instructions below assume that the downloaded jar file (jolokia-jvm-1.6.0-agent.jar) is placed under the location '/opt/kafka/libs/'.

### Kafka Brokers

To configure Kafka Brokers to expose the Jolokia API, you can add the following in <KAFKA_HOME>/bin/kafka-server-start.sh, just before the 'kafka-run-class.sh' call:

```
export JMX_PORT=9999
export RMI_HOSTNAME=`hostname -I`
export KAFKA_JMX_OPTS="-javaagent:/opt/kafka/libs/jolokia-jvm-1.6.0-
agent.jar=port=8778,host=0.0.0.0
-Dcom.sun.management.jmxremote.password.file=/opt/kafka/config/jmxremote.p
assword -Dcom.sun.management.jmxremote.ssl=false
-Djava.rmi.server.hostname=$RMI_HOSTNAME
-Dcom.sun.management.jmxremote.rmi.port=$JMX_PORT"
```

Note that example above is using 'hostname -I' to setup the 'RMI_HOSTNAME' environment variable. In multiple IP machines, this will need to be tweaked to gather the IP you care about for RMI connections.

You can choose a different port for JMX (9999 above) and Jolokia (8778). If you have an internal IP to lock Jolokia onto you can replace the "catch all" 0.0.0.0 by your own IP. Notice this IP needs to be accessible from the telegraf plugin. You can use the option '-Dcom.sun.management.jmxremote.authenticate=false' if you don't want to authenticate. Use at your own risk.

## Objects and Counters

The following objects and their counters are collected:

| Object: | Identifiers: | Attributes: | Datapoints: |
|---|---|---|---|
| Kafka Broker | Cluster<br>Namespace<br>Broker | Node Name<br>Node IP | Replica Manager Fetcher Max Lag<br>Zookeeper Client Connections<br>Zookeeper Client Connections (15m rate)<br>Zookeeper Client Connections (5m rate)<br>Zookeeper Client Connections (mean rate)<br>Zookeeper Client Connections (1m rate)<br>Replica Manager Partition Count<br>Thread Count Daemon<br>Thread Count Peak<br>Thread Count Current<br>Thread Count Total Started<br>Offline Partitions<br>Produce Requests Total Time (50th Percentile)<br>Produce Requests Total Time (75th Percentile)<br>Produce Requests Total Time (95th Percentile)<br>Produce Requests Total Time (98 Percentile)<br>Produce Requests Total Time (999th Percentile)<br>Produce Requests Total Time (99th Percentile)<br>Produce Requests Total Time<br>Produce Requests Total Time Max<br>Produce Requests Total Time Mean<br>Produce Requests Total Time Min<br>Produce Requests Total Time Stddev<br>Replica Manager ISR Shrinks<br>Replica Manager ISR Shrinks (15m rate)<br>Replica Manager ISR Shrinks (5m rate)<br>Replica Manager ISR Shrinks (mean rate)<br>Replica Manager ISR Shrinks (1m rate)<br>Request Handler Avg Idle<br>Request Handler Avg Idle |

# Troubleshooting

Additional information may be found from the Support page.

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